Engineering Mechanics 1 · Unit_04 · UNIT 4

DRAFT — UE26ME142A · Unit_04 · UNIT 4 · CLASS 40 Moment of Inertia Numerical

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · CLASS 40 Moment of Inertia Numerical
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

CLASS 40 Moment of Inertia Numerical

1 CLASS 40 Moment of Inertia Numerical

Study focus: CLASS 40 Moment of Inertia_ Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000042 · EV0000043

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

This is the first problem in today's class A slash 35.

Key details

- Determine the moment of inertia about the x-axis of the square area without and with the central circular pole.
- We have been asked to find moment of initiative this square about this x axis in the absence of this circular pole, and second.
- So the first part is in the absence of this, what is the moment of inertia of this square about the x-axis.
- If you look at this x axis, this x axis is.
- Nothing but centroidal x-axis because this square is symmetrical about this x-axis.
- That's why this itself will be the centroidal x-axis.
- Then how to find moment of inertia phase square about its centroidal x axis?
- We know we can make use of the formula of a rectangle itself.
- The moment of inertia of a rectangle about its centroidal x-axis is BH cubed by 12.
- four R and we can find IX.
- Square ear, x axis is here.
- This is without considering the central circular hole.
- What is B here?
- 4R and H is also 4R.
- So next I bar X will be equal to in this case it will be 6.
- 4 r to the power of 4 by 12.
- Simplify this, we are going to get it as 21.33 into r to the power of 4.
- This is the moment of inertia of this.

- Square about the x axis.
- About X axis that will be equal to moment of inertia of S square.
- About x axis minus moment of inertia of a circle above the same x axis.
- Pi r to the power of 4 by 4 because this itself is the centroidal x-axis even for circle.
- Cube by 12 minus pi into here radius is R itself.
- Sorry, R to the power of 4 by 4.
- This is the moment of inertia of the square with the central circular hole.

Example or application from the lecture

- They have given a square and a side 4R.
- We have to consider the circular hole here, and we have to find the moment of inertia of only the shaded area.
- Let's start with the solution.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

Core explanation · 05:00–09:59

Then this is the next question A slash 39.

Key details

- Calculate the polar radius of gallation of the shaded area about the center O of the larger circle.
- Look at this.
- There is a circle of radius 200 millimeter, and this is the center of the larger circle.
- Next, there is one more smaller circle.
- And C is the center of this smaller circle of radius, 100 mm.
- And if you look at these two O and C, they are on a same horizontal line.
- Here, they have asked us to find polar radius of gyration.
- Polar radius of gyration is nothing but Kz.
- Then how to find Kz?
- To find Kz, we have two formulas.
- One is under root of IZ by A.
- Where iz is polar moment of inertia and capital A is area.
- Then there's one more formula that is Kz is equals to under root of.
- Kx squar plus Ky squared.
- This formula we are going to use when we know the radius of variation about x axis and radius of variation about y-axis.
- Here, since we don't know the radius of variation about x-axis and y axis, we are going to use the first formula.
- That is Kz is equals to under root of IZ by A.
- Next, as I said before, iz is the polar moment of inertia.
- Since here we are talking about point O, that polar moment of inertia should be about the point O.

- Next, how to find polar moment of inertia of the shaded area about point about this point.
- Here, we want only the shaded area.
- Polar moment of inertia of this smaller circle about the point O.
- Then we are going to get the polar moment of inertia of the shaded area.
- Then find only the shaded area, then use the formula iz by A.
- We need to find Kz, that is polar radius of garation of the shaded area about 0.0.
- That is polar radius of garration about point.
- That is equals to under root of I z that is polar moment of inertia about the point.
- [ASR low confidence; check audio].
- Divided by total area capital E.
- Then how to find this polar moment of inertia that is Iz about point four.
- Polar moment of inertia about 0.4 will be equals to.
- I said about.
- O will be equals to polar moment of inertia of a bigger circle that is 200 mm radius circle about point O minus.
- Polar moment of inertia of a smaller circle that is 100 mm circle about the same point.
- Next, what is polar moment of inertia of this bigger circle?
- That is 200 mm circle, about 0.
- If you observe the figure carefully, that o itself is the center of the bigger circle.
- Polar moment of inertia about that point O will be equals to I bar X plus I bar Y.
- 4 plus I bar y is pi r to the power of 4 by 4.
- And what is radius?
- Radius is 200.

Example or application from the lecture

- Let's get into the solution.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000046 · EV0000047 · EV0000048 · +1 source units in pack.json

Core explanation · 10:00–14:59

200 to the power of 4 by 4 plus pi into 200 to the power of 4 by 4.

Key details

- Once we simplify this, we are going to get the value as 8 pi into 10 power 8 mm to the power of 4.
- This is the polar moment of inertia of a bigger circle about the point.
- So we are done with the first term.
- We need to find next.
- Polar moment of inertia of a smaller circle above the point O, that is Iz.
- I said of 100 mm circle about point O.
- This will be equals to here.
- Do we have a standard formula for polar moment of inertia about point O?
- No, we don't have standard formula, so we have to apply the parallel axis theorem.

- Remember, one of the axis is passing through O, that is Z-axis passing through O, and one more Z-axis is passing through C.
- The Z-axis passing through C is the centroidal z-axis of the smaller circle.
- Then what is i bar z?
- We know for a circle i bar z is pi r to the power of 4 by 4 plus pi r to the power of 4 by 4.
- Then what is distance?
- Distance is the distance between O and C, that is 15.
- Here, r is 100, so pi into 100 to the power of 4 by 4, plus pi into 100 to the power of 4 by 4, plus pi into 100 square into 50 square.
- Just use the calculator and find the value.
- We are going to get this as these two terms will be equal to 15 pi into 10 power 6.
- Then plus pi r square, that is pi into hundred square into fifty square, will get it as twenty-five pi.
- into 10 power 6.
- Then add these two, it will be 75 pi into 10 power 6 mm to the power of 4.
- This is the moment of inertia of a smaller circle about the point O.
- Next, I said of the shaded area.
- This will be equal to moment of inertia of the bigger circle, that is 8 pi into 10 power 8 minus moment of inertia of the smaller circle, that is 57 pi into 10 po 6.
- Seven, seven, five into ten per eight, mm to the power of four.
- Next, we should know the value of area.
- Here, area is area of the shaded region.
- That area will be equals to A is equals to.
- Pi r square minus pi r square.
- R square is 200 square minus 100 square.
- eight and a as thirty thousand into five and solve it we are going to get the value of K said as.

Example or application from the lecture

- Then plus area we have to consider here area is phi r square.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

Core explanation · 15:00–19:59

One five point four five millimeter.

Key details

- This is the radius of variation of the shaded area about the point O.
- That's it for the today's class.
- If you come across any queries, you can post your queries in PESO forum.

Example or application from the lecture

- Try to solve the problems given in the assignment as well as in the question bank.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000054 · EV0000055 · EV0000056 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
CLASS 40 Moment of Inertia_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/050_CLASS_40_Moment_of_Inertia__Numerical__0e23de7b877dd375.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 40 Moment of Inertia_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/050_CLASS_40_Moment_of_Inertia__Numerical__0e23de7b877dd375.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 40 Moment of Inertia_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/050_CLASS_40_Moment_of_Inertia__Numerical__0e23de7b877dd375.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9984; only the JSON segmentation supplies the displayed timestamps.
CLASS 40 Moment of Inertia_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/050_CLASS_40_Moment_of_Inertia__Numerical__0e23de7b877dd375.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md have no readable `.slides.md` text; their original page images are included.
3. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
4. 4 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
5. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
6. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
7. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
8. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00014	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0065. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00015	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0065. Sidecar variants compare...
SRC00016	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0065. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00017	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0065. Sidecar variants compare...

C Revision and retrieval practice